An Information-Theoretic Approach to Nanopore Sequencing for DNA Storage

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***Abstract*—DNA storage is a cutting-edge technology that has the potential to store all of the promised information of the future, which approximately doubles each year, without erasing the information of today. A core process of such systems is the reading process, i.e., the sequencing of DNA, which is yet to be fully explored in terms of information-theoretic techniques. This article is an exposition of current day research of the nanopore sequencer from the perspective of an information theorist who wishes to reach its capacity in the context of DNA storage systems. The raw data signal from state-of-the-art nanopore sequencers is traditionally modeled together with algorithms based on neural networks, whose obscurity makes it difficult to develop efficient and reliable coding schemes. Alternatively, the nanopore sequencer can be approximated by simple yet tractable models that capture its essence, for which information-theoretic techniques can be applied. Different nanopore models from the literature are explored, including their connection with noisy duplication channels. The open problems are highlighted, and how their solutions could improve nanopore-based DNA storage systems.**

I. Introduction

Information has approximately doubled every two years since the digital revolution and is expected to reach the order of zettabytes in the coming decades [1]. DNA is a medium that can, in principle, store zettabytes in a chemical solution that weighs just a few grams, which is in stark contrast to the large data centers in use today. If one is to store all of the promised information of the future without erasing the information of today, the miniaturization of our data storage systems is an inevitability. Richard P. Feynman, the father of nanotechnology, envisioned such a future back in 1959 when he gave a talk ("There's plenty of room at the bottom", [2]) entertaining the following question:

> "Why cannot we write the entire 24 volumes of the Encyclopedia Britannica on the head of a pin?"

In 2012, significant progress towards achieving Feynman's vision was taken in [3], where around a megabyte was stored in DNA, almost the size of one volume of the Encyclopedia Britannica. In 2018, this was taken the furthest to date in [4] where over 200 megabytes were stored in DNA, more than 24 volumes of the Encyclopedia Britannica, and recovered without error.

Current-day DNA storage systems can be simplified down to six processes: encoding, synthesis, storage, polymerase chain reaction (PCR), sequencing, and decoding. If we view the DNA channel as the synthesis, storage, PCR, and sequencing processes, then the role of the encoding process is to make our stored information robust to distortions from the channel processes so that the decoding process can recover the information without error. However, before designing the encoder and decoder (i.e., the channel code), we must have an accurate channel model for the channel processes that are introducing errors. The "DNA channel" commonly refers to the process of synthesizing information into many short strands of DNA that are copied, shuffled, and corrupted by substitutions, insertions, and deletions. The information-theoretic limits of variations of this channel model have been studied in recent years [5]–[7], and simultaneously codes to approach these limits have been proposed (e.g., [8]–[13]). However, the DNA channels considered in the literature today tend to hide away the complexity of the constituent processes of the channel: this is certainly an advantage for theoretical analysis, but it reduces our ability to provide tight performance limits.

This article will focus on the sequencing process, which is the most critical contributor to DNA channel impairments. In particular, we consider more accurate models for the new generation DNA sequencer by Oxford Nanopore Technologies (ONT) [14], which promises to expand the current limits of DNA storage, thanks to its reduced cost of operation and its portability that enables operation outside a lab.

The primary task of a DNA sequencer is to read a DNA strand composed of base symbols and output a signal that contains as much information about the encoded data as possible. The typical approach to DNA sequencing has been to decode this raw signal using a neural network (NN) "basecaller" algorithm, which places the sequencing process in a black box. This approach was carried over to nanopore sequencing for DNA storage, resulting in methods that do not fully utilise the raw signal that is rich in information, in favor

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of the processed output from the basecaller. Alternatively, we can use our understanding of the nanopore sequencing process to describe the sequencer with a tractable model. The purpose of this article is to explore the nanopore sequencing process and its different models in the literature to describe it.

In Section II, the nanopore sequencing process is explored. This gives insight into how the encoded information is modulated and distorted by the sequencer. In particular, we observe that the nanopore channel belongs to the class of noisy duplication processes. In Section III, we discuss different software packages that attempt to simulate the nanopore sequencing process, which have varied in approach over the years. For example, we introduce the Scrappie [14] technology demonstrator released by ONT. This highlights features of the raw signal that can be leveraged for modelling Oxford nanopores . In Section IV, we introduce the noisy nanopore channel (NNC) from [15] that models the key features of the raw signal generated by Scrappie. In essence, this is the simplest noisy duplication channel that accurately describes the nanopore channel. In Section V, we explore a modelling approach that includes an equalization step to process the data (e.g., the basecaller). There are two significant channel models from the literature that follow this approach: the simplified nanopore channel (SNC) [16], for which coding for nanopore sequencers was first considered in an information-theoretic setting, and more recently the substitution-insertion-deletion (SID) channel with memory [17], [18]. In Section VI, the decoding problem is considered for channels with noisy sample duplications. This discussion is centred around segmentation algorithms, which naturally leads to the dynamic time-warping (DTW) algorithm that is relevant to both decoding and code design. In Section VII, we conclude the article by summarizing the open problems that could be interesting to the information theory community.

II. Nanopore Sequencing

The nucleotides are loosely labelled as the *bases* of the quaternary alphabet $\mathcal{B} = \{\mathsf{A}, \mathsf{T}, \mathsf{C}, \mathsf{G}\}$ with the chemical structure of DNA shown in Fig. 1. In nanopore sequencing, the nanopore is a microscopic pore that is used as a biosensor to detect *nucleotide* molecules that compose a strand of single-stranded DNA (ssDNA), schematically illustrated in Fig. 2. The nanopore is a passageway (red in Fig. 2) for incoming nucleotides (blue in Fig. 2) that creates a tunnel through a membrane (gray in Fig. 2) that allows the flow of an electrical current of ions along with the ssDNA strand. Modern nanopore sequencing devices typically contain one or more arrays of nanopores, each with their own unique current signal.

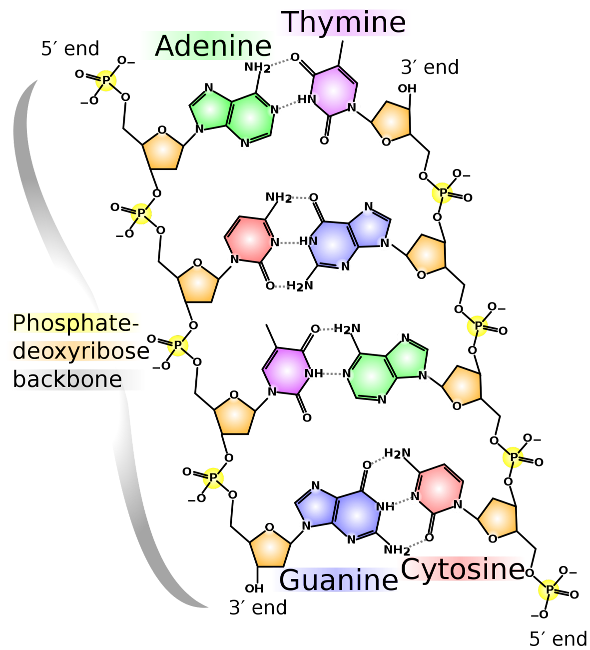

Fig. 1. Chemical structure of the nucleotide molecules that compose double stranded DNA. The phosphate backbone is the support for the four bases Adenine (A), Thymine (T), Cytosine (C), and Guanine (G) that ligate through two or three hydrogen bonds in pairs A-T and G-C (Image credit: Wikipedia).

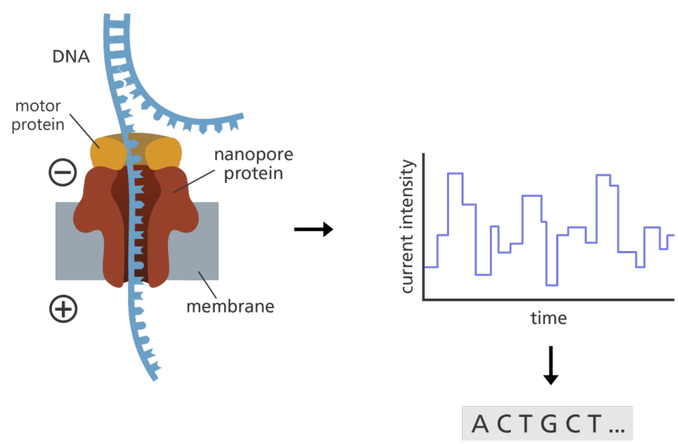

Fig. 2. Overview of the nanopore sequencing process (Image credit: Genome Research Limited). The nanopore (in red) gives passage to a single strand DNA (in blue). The nucleotides are driven through the nanopore by the motor protein (in yellow). The voltage across the membrane (in gray) generates an electrical ionic current that is modulated by the nucleotides inside the nanopore. The current signal in the pA scale carries the information about the nucleotides traslocating the nanopore.

This signal can be measured for each nanopore in a given array, enabling multiple reads in parallel. The chemical properties of each base cause a unique disturbance to the current signal, yielding a measured current signal that can be used for detection. Without intervention, the nucleotides traverse the nanopore so quickly that their signal-to-noise ratio would be too low for accurate detection. This is addressed by equipping the nanopore with a motor protein (yellow in Fig. 2) that attempts to control the speed of the nucleotides, ideally in a ratchet-like fashion that provides enough time to track fluctuations of the current. Despite all of this, the state-of-the-art nanopore sequencers by ONT [14] are still limited by distortions caused by the high variability of the molecular interactions in the process.

While the goal of nanopore sequencing is to detect a single base, the reality is that there are at least a few nucleotides inside the nanopore at a given time that influence the measured current signal. Therefore, the detection of a single base is distorted by neighbouring nucleotides inside the pore. Additionally, the contribution of each base to the average current measurement is not uniform, since the width of the nanopore varies, with its narrowest point close to the center. The magnitude of the current measurement is determined by how severely the bases in the pore cause a blockage, and hence we intuitively conclude that the center-most base has the greatest opportunity to influence the current measurement. This results in the center-most bases greatly determining the measurements, especially the individual base that is travelling through the narrowest section. While the center-most base does not correspond to the last input base, it is often convenient to consider it as being so since we intuitively expect every input to have a greater influence than the bases before it.

The width of the nanopore allows for easier blockages, which we just saw depends on the position in the nanopore, but the molecular weight of each base in the pore also determines the severity of the blockage. It turns out that there are two groups of bases that have similar blockages, which are the *purine* (A and G) and *pyrimidine* (T and C) chemical groups. The distinguishing property between these two groups is their molecular weight, measured in grams per mol, which is relatively larger for the purines that yield significantly smaller magnitudes of current measurements. Hence, the single-most distinguishing feature of the bases in the nanopore is the chemical group of the center-most base, but there is still a significant amount of variation in current measurements caused by the surrounding bases that cannot be ignored.

The motor protein that drives the nucleotides through the pore is also not ideal, since it cannot do the ratcheting mechanism at consistent time intervals. In practice, nucleotides are ratcheted into the nanopore at random times, and since measurements of the electrical current are deterministically clocked, this produces a random number of samples for each unique and fixed-length sequence of nucleotides that pass through the nanopore. Additionally, the speed of the motor protein is affected by the bases that pass through it, although not in the same way as the current measurements. As discussed earlier, the current measurements are affected by the bases in the pore, however, while it is plausible that different molecular weights of the bases could cause a variation in friction that alters the speed, there is more evidence to suggest that only the bases inside the motor protein cause the predominant change in speed. In Fig. 2, it can be seen that the motor protein is located at the entrance of the nanopore, suggesting that the bases that affect the traversal speed are different to those that cause the current blockage.

In addition, each sample is corrupted by what we will refer to as "measurement noise". This includes at least two noise processes. The first is thermal noise that is common to many communications systems, and the second is the chemical noise. The chemical noise can be broadly described as the noise that occurs due to random local concentration variations of the molecules in the pore. Due to variability in nanopore structure, the average variation within each pore could also vary, causing a variable fading coefficient. This results in average current levels that depend on the nanopore in use and its operating conditions at a given time instant.

III. Simulation and Nanopore Modelling

While the Oxford nanopore sequencer is very accessible and relatively inexpensive to operate, its ease-of-use is limited by the recipe of chemical processes that must be executed before the sequencer can be used. In addition, the cost of synthesizing strands of DNA in the first place is still significant today. Hence, it would be convenient to have a virtual simulator of the nanopore sequencer that can be probed without any domain-specific knowledge or cost. This is where simulators of the nanopore sequencer come into play.

The simulators that have been released for the nanopore sequencer can be divided into two categories. Category I includes the simulators that are based on modeling a simple statistic of the nanopore sequencer, such as the rate of substitutions, insertions, and deletions after decoding the raw data with an NN that estimates the base sequence (often referred to as "basecalling"). Category II includes the simulators that are based on

TABLE I
STATIONARITY OF SCRAPPIE FOR MEMORY τ

Stationarity	τ				
	1	2	3	4	5
$\Delta_\mu(\tau)$	0.45	0.38	0.22	0.21	0.08
$\Delta_\sigma(\tau)$	0.03	0.03	0.01	0.01	0.01

modeling the raw current signal (e.g., Scrappie [14] and DeepSimulator [19]), before basecalling, which means modeling the current measurement levels and the number of the samples measured for each base input. Category II is clearly an improvement on the first category since it does not hide away the complexities of the nanopore sequencer, and instead tries to model them. This distinction was behind the early development of DeepSimulator, whose authors argued that [19]:

> "the current signal is the essence of nanopore sequencing, yet there is no such simulator that attempts to mimic the signal generation step."

However, such simulators require learning many parameters that describe the average current levels and the number of samples, which is impeded by the inability to perfectly synchronize the current levels. Given the model complexity, these parameters are typically captured by an NN with memory across a long base sub-sequence (e.g., forty bases) [14], [19]. Alternatively, there are "pore models" that model these parameters with a memory of just a few τ bases (e.g., [20], [21]).

The Scrappie simulator. In 2017, the Scrappie simulator was released by ONT as a technology demonstrator for their sequencers, which is essentially an NN that was trained on genomic data to extract characteristic features of the current signal. Scrappie outputs triplets of statistical parameters that model: (1) the average current levels, (2) the measurement noise in the current levels, and (3) the duration of the current levels. The output of the nanopore sequencer can be simulated by sampling from a distribution using these three parameters, which means sampling from a Gaussian (or Laplacian, as recommended by ONT in their documentation [22]) distribution with mean equal to the level parameter, μ_ℓ, variance equal to the noise parameter (squared), σ_ℓ^2, repeated for a random number of samples, d_ℓ. Although there are just three output parameters per input base, they obey a complicated relationship due to memory that evolves with input index ℓ depending on the bases that came before it.

The pore model. The key property of pore models is that there are only a finite number of bases inside the pore at a given time that contribute to the memory in the nanopore channel, which means the output parameters of Scrappie should only depend on a finite number of previously input bases. We can deduce how many bases this is by assuming different memory lengths and calculating their *stationarity*, which we define using the maximum standard deviations

$$\Delta_\mu(\tau) = \max_{b_{\ell-\tau+1}^{\ell}} \mathsf{SD}[\mu_\ell | b_{\ell-\tau+1}^{\ell}]$$

$$\Delta_\sigma(\tau) = \max_{b_{\ell-\tau+1}^{\ell}} \mathsf{SD}[\sigma_\ell | b_{\ell-\tau+1}^{\ell}]$$

for a given memory length τ. This is shown in Table I by measuring these variances with equal weightings for

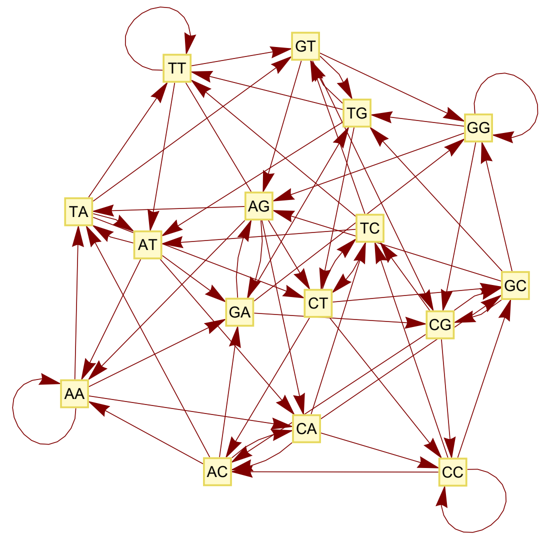

Fig. 3. The state-space of a two-base pore model with measurement levels from Scrappie [22] in Table II.

TABLE II
AVG. MEASUREMENT LEVELS FOR TWO-BASE SCRAPPIE MODEL.

	B_1B_2 *	$f(B_1B_2)$ †
B_2 is purine	AA	-0.64
	TA	-0.25
	CA	-0.36
	GA	-1.06
	AG	-1.08
	TG	-0.55
	CG	-0.78
	GG	-1.34
B_2 is pyrimidine	AT	1.23
	TT	0.74
	CT	1.04
	GT	0.92
	AC	0.61
	TC	0.38
	CC	0.37
	GC	0.47

* Base B_2 corresponds to the base closest to the center of pore.
† Median-normalized current levels from Scrappie [14].

all input bases, demonstrating that $\tau = 5$ allows a reasonable stationarity assumption relative to the Scrappie model. The durations are not included since they depend on a different sub-sequence of bases at the entrance of the nanopore, as discussed in the previous section, but a similar analysis can be done. Once τ is chosen, the NN model can be simply approximated by a pore model with memory τ. For example, if $\tau = 2$ then the pore model approximating Scrappie is described by the state space in Fig. 3 with level estimates in Table II. This is only for illustration purposes, but of course its stationarity $\Delta_\mu(2) = 0.38$ makes it a very poor choice in practice.

Note that the property regarding purine and pyrimidine chemical groups discussed in the previous section is verified in Table II, where the base closest to the center, B_2, produces negative values if it is a purine, and positive values if it is a pyrimidine.

The parameter estimation problem. Given the reliance on having an accurate pore model to model the nanopore sequencer, there remains the question on whether it is theoretically possible to estimate its parameters. If every current level had a fixed duration, then estimating the pore model would be as simple as taking a sample average. However, the random number of samples produced by the sequencer de-synchronizes the current levels such that we can never be sure where one symbol starts and ends in the current signal. Hence, an interesting open problem is to describe exactly how the pore model can be estimated, e.g., for a long sequence of noisy levels with sample duplications, can the levels be estimated perfectly like in the synchronized case? Alignment algorithms have been developed to solve this problem in practice using multiple reads of the same DNA strand, e.g., [23] discussed these challenges and proposed the QAlign algorithm.

This problem can be mitigated increasing by the number of samples per current level, which makes it far more reliable to estimate the levels than if they were in the range of say one to two samples.[1] In addition, it may be possible to increase the sampling rate of the sequencer so that this consideration is effectively eliminated.

IV. The Noisy Nanopore Channel

The nanopore sequencer is a device with complicated dynamics, influenced by many factors including room temperature and pore construction, and hence there are seemingly many parameters to be considered for an accurate model of the nanopore sequencer. While traditional approaches based on NNs may yield the best outright model of the nanopore, it does not offer much insight into its fundamental structure that could be further understood with information-theoretic techniques. The first channel model that we will introduce is based on the nanopore sequencing process given in Section II, which was indirectly defined as some kind of noisy duplication process. From the class of noisy duplication channels, we explore a channel (illustrated in Fig. 4) with i.i.d. duplications, inter-symbol inference, and AWGN. This is motivated by the simulators discussed in Section III.

Mapping bases to states. For m channel-uses, the input to the nanopore channel is the sequence of bases $\{B_n\}_{n=1}^{m+\tau-1}$, however, we can instead do a one-to-one mapping of these input bases to a sequence of states $\{S_\ell\}_{\ell=1}^{m}$, where each state $S_\ell = (B_{\ell-\tau+1}, B_{\ell-1}, \ldots, B_\ell)$ describes the sub-sequence of bases that are inside the nanopore after having input the ℓ-th base ("Base-to-state Mapping" in Fig. 4). Hence, we can think about the input to the NNC as being a first-order Markov chain over a state-space Ω of all the possible τ-tuples of bases in the pore. We note that this notation is different from the traditional shift-register model that would define the state as the previous $\tau - 1$ bases, but in this application, it makes more sense to choose the state as being all the bases in the nanopore at a given time.

The Markov chain can be visualized as a graph (e.g., Fig. 3) of nodes from state-space Ω, the state-space of the nanopore, and edges from $\{\mathsf{A}, \mathsf{T}, \mathsf{C}, \mathsf{G}\}$, the input to the nanopore; for an initial state S_0 (node 1), the next state S (node 2) may be achieved by an input base B (edge between node 1 and node 2). The graph representation is often very useful for describing the nanopore, especially when we later discuss source constraints.

Sample duplications. The first stage of the NNC is a duplication channel that models the sample duplications, with a Markov source input that models the memory in the nanopore ("Duplication Channel" in Fig. 4). We refer to *sample* duplications to distinguish from the duplication of bases, which does not physically occur in the nanopore. An incoming base produces a state transition and the number of duplicated samples decides how long we remain in that state. This determines the number of samples that we take from the raw signal for each channel use.

The input to the duplication channel will be the Markov states $\{S_\ell\}_{\ell=1}^{m}$, and the duplication channel duplicates the ℓ-th state K_ℓ times, producing the variable-length output sequence $\{Z_t\}_{t=1}^{T_m}$, where $T_m = \sum_{\ell \le m} K_\ell$ is the total length after duplications. For convenience, it is useful to reparameterise the sequence of sample duplications $\{K_\ell\}_{\ell=1}^{m}$ into the so-called *segmentation points* (or *jump times*) $\{T_\ell\}_{\ell=1}^{m}$, defined as $T_\ell = \sum_{i \le \ell} K_i$ such

[1] For the Oxford nanopore sequencer the number of samples per current level varies between 5 and 15.

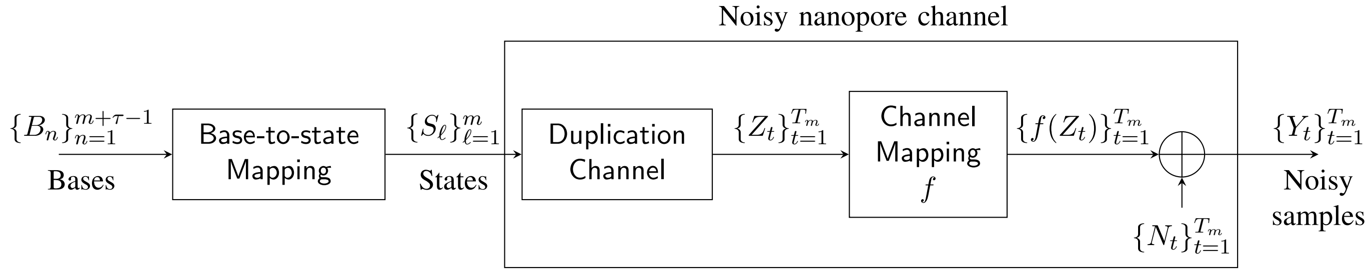

Fig. 4. Block diagram of the noisy nanopore channel [15] for modeling the nanopore sequencer.

that they correspond to the index of the last duplicated sample in the output sequence for the ℓ-th input. Using these jump times, the output of the sub-channel after duplication is

$$\underbrace{Z_1, Z_2, \ldots, Z_{T_1}}_{K_1}, \underbrace{Z_{T_1+1}, \ldots, Z_{T_2}}_{K_2}, \ldots, \underbrace{Z_{T_{m-1}+1}, \ldots, Z_{T_m}}_{K_m}$$

where each segment labelled with an underbrace corresponds to a sequence of identical duplicated states that were produced from a single input state.

Although the input to the duplication channel is a Markov chain that follows the Markov property between each consecutive pair of states, the output only satisfies the Markov property at the jumps, i.e., going from Z_{T_ℓ} to $Z_{T_\ell+1}$. The output of this sub-channel is known as a *semi-Markov chain* [24]. There is a body of research on semi-Markov chains that we can draw from, giving us a convenient framework to concisely describe and analyze the NNC. Notably, the *asymptotic equipartition property* has been shown to hold for the entropy rate of semi-Markov chains [25].

Inter-symbol interference and AWGN. The second stage of the NNC that models inter-symbol interference and measurement noise ("Channel Mapping" and the "addition" in Fig. 4). The input to this stage is the semi-Markov chain output of the first stage. The task of this stage is to map each state Z_t into a real-valued output sample $f(Z_t)$, where f is the so-called channel mapping (identical to the "pore model" in Section III) that describes the inter-symbol interference in the channel. Properties of this mapping were considered in [26].

The remaining distortion to include is the measurement noise, which we choose to be AWGN such that the output sample at time t is $Y_t = f(Z_t) + N_t$ for Gaussian-distribution noise $N_t \sim \mathcal{N}(0, \sigma^2)$. Applying this mapping for all T_m states in the semi-Markov chain, we obtain the output sequence of the NNC as $\{Y_t\}_{t=1}^{T_m}$, which is a *hidden* semi-Markov chain.

The cascade of the two aforementioned stages forms the NNC in Fig. 4 that was proposed in [15]. Compared to the Scrappie simulator, there are two key simplifications made. The first is that the durations are i.i.d.: while the channel can be naturally extended to non-i.i.d. durations, current-day models of this do not seem accurate enough to be used with confidence. The second is the finite state-space to model inter-symbol inference, allowing the channel parameters to be represented as a look-up table (e.g., Table II), rather than the NN used by Scrappie. These approximations enable us to emulate the physical behavior of the nanopore with a tractable model that can be analyzed using information-theoretic techniques.

Related channels. While the nanopore channel could be more closely modeled by a generalized NNC, these kinds of channels are relatively unexplored and its special cases are challenging in their own right:

- *Finite-state channel*: If the segments of output samples for each input are synchronized, e.g., by using markers, then we have a finite-state channel. This channel has been extensively studied, and some of its techniques may be adopted to this new scenario.
- *i.i.d. binary-input noisy duplication channel*: If the input is i.i.d. and has a binary alphabet, then we arrive at the simplest noisy duplication channel. Only tangentially related results for this channel exist (e.g., [15], [27]).
- *Sticky channel*: If we remove the noise in the channel, i.e., setting $\sigma^2 = 0$, then we end up with a sticky channel (or duplication channel) with memory. This channel has been studied extensively in the memoryless case (e.g., [28]). Results in this scenario could be important to noisy duplication channels in the low-noise regime.

From these special cases, we can further appreciate the challenging open problems that face noisy duplication channels. Some recent results have been proposed using ideas from finite-state channels, including MAP decoding algorithms and information rate estimates [15], but many problems remain. The nanopore application could help to motivate new results in this area.

V. Exploring the Equalization Approach to Nanopore Modelling

The approach taken in the previous section was to model the nanopore channel at the sample-level as accurately as possible under the constraint of model tractability. The advantage of this approach is that we minimize degradation of nanopore channel by using a channel that closely models it. However, this method has a disadvantage in that it necessitates acquiring a significant number of channel parameters to characterize it. An alternative approach is to concatenate a channel equalization step with the complex model of many parameters to create an equalized model of fewer parameters that are readily known. In this section, we explore two significant channel models from the literature that are based on equalization to better understand their strengths and weaknesses in practice.

A. Simplified nanopore channel

Various models of the nanopore channel have been proposed since the popularization of nanopore sequencing in the early 2010s. The earliest model to be considered in an information-theoretic setting was the "simplified nanopore channel" (SNC) introduced and analyzed in the foundational work of [16] from 2018, whose principal contributions were two-fold: describing a tractable model of the nanopore sequencer in terms of the pore model, and advancing the theory of deletion channels to this new scenario.

Backtracking and skippings. As discussed in Section II on nanopore sequencing, [16] comprehensively outlined the primary distortions of the nanopore sequencer: inter-symbol interference (channel memory), random dwelling time (sample duplications), level fading, and noisy observations. In addition, "backtracking" and "skipping" synchronization errors were considered, better known as base-insertions and base-deletions, respectively, which were not included in our earlier description of the nanopore sequencer. These distortions were attributed to malfunctions of the motor protein that cause backtracking and mis-stepping of the DNA strand in the pore. This consideration was likely due to the hypotheses of their collaborators in [21] based on experimental data from early nanopores in 2014. To the best of our knowledge, it remains unclear if these hypotheses are true for Oxford nanopores that are in common use today. In any case, many novelties of the channel model in [16] are still important for modelling the nanopores of today.

Channel equalization. The simplified nanopore channel includes an equalization step that processes the nanopore channel output $\{Y_t\}_{t=1}^{T_m}$ with a level finding algorithm that gives $\hat{m}$ level estimates $\{\hat{X}_\ell\}_{\ell=1}^{\hat{m}}$. A specific level finding algorithm is not given, but rather it refers to one of many possible change-point algorithms [29] that attempts to locate transitions in a time series. Since this model is based on a high sampling rate of roughly 50kHz for segment durations of >40ms on average (2000 samples per segment on average), it was observed that estimation of each $\hat{X}_\ell$ was very accurate and therefore decided to assume the level finding is perfect. That is, we have $\hat{X}_\ell = X_\ell$ for all $\ell \in \{1, 2, \ldots, m\}$. Insertions and/or deletions occur when $\hat{m} \neq m$, which we know is possible for such algorithms based on sub-optimal segmentation, and are included with the modelling of backtracking and skippings. The channel equalization stage is completed after discretization of the level estimates $\{\hat{X}_\ell\}_{\ell=1}^{\hat{m}}$ into discretized levels $\{V_\ell\}_{\ell=1}^{\hat{m}}$, and now the SNC in Fig. 5 is ready to be defined as follows.

Channel definition. The input to the channel is the sequence of bases $\{B_n\}_{n=1}^{m+\tau-1}$, which undergo the base-to-state mapping from Fig. 4 to give the sequence of m states $\{S_\ell\}_{\ell=1}^{m}$. As earlier, these states represent the bases in the pore and are again chosen to form a Markov chain. However, since we are modelling the nanopore channel combined with the aforementioned level finding and discretization algorithms, the next step is not to model sample duplications but rather to model deletions of bases (or equivalently, states). The subsequent deletion channel ("Deletion Channel" in Fig. 5) deletes the indices from a random set of indices $\mathcal{D}$, chosen according to the i.i.d. deletion rate p_d for each ℓ-th state. The remaining states are the corrupted states $\{S_\ell\}_{\ell \notin \mathcal{D}}$, which is a *watched* Markov chain (or a *censored* Markov chain) [30] that is like a classical Markov chain except that we only observe a subsequence of states. That is, we have a new state space $\Omega' = \Omega \times \{\mathsf{NoErr}, \mathsf{Del}\}$ and we only observe the subset of states $(s, e) \in \Omega'$ with $e = \mathsf{NoErr}$ (no error) and do not observe the states with $e = \mathsf{Del}$ (deletion). The final step of the SNC is to map the remaining states to levels ("DMC (Fading)" in Fig. 5), which could be done using mapping using the channel mapping f from Fig. 4, but the model instead considers fading noise in these levels. We will shortly discuss this in detail, but it turns out that the level fading in the SNC combined with discretization of the levels is a discrete memoryless channel (DMC). Thus, the output of the channel model are the discretized levels $\{V_\ell\}_{\ell=1}^{m-|\mathcal{D}|}$ that form a hidden watched Markov chain.

Level fading. In practice, the channel mapping f varies over time due to various factors, from varying conditions of the nanopore to noise in the channel mapping itself as a result of imperfect parameter estimation (recall from Section III that the choice τ caused variations in

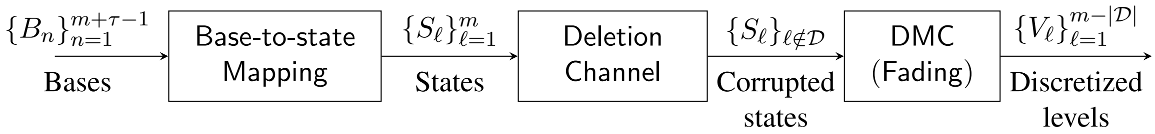

Fig. 5. Block diagram of the simplified nanopore channel [16] for modeling the nanopore sequencer.

f due to non-stationarity). This is modelled in the SNC by mapping every state $s \in \Omega$ to a uniformly random level in the interval $[f(s) - \Delta, f(s) + \Delta]$, where Δ is the worst-case deviation from the mean. This results in a collection of intervals that have a lot intersections, inducing a partition on the continuous range of current levels. To include level discretization, we use a discrete partition in place of the continuous partition. Such a partition forms a set of sufficient statistics for the channel output of a DMC.

B. Substitution-insertion-deletion channel with memory

There are numerous studies in the literature of nanopore sequencing that analyze the rate of substitution, insertion, and deletions of bases after basecalling (e.g., [31]). This is common in the genomic sequencing community that, independently of DNA storage, develop basecallers with the objective of reducing the rate of these synchronization error events in the basecalled sequence. Hence, it is tempting for an information-theorist to bring the fruits of these works to DNA storage by applying the substitution-insertion-deletion (SID) channel. This is equivalent to the equalized nanopore channel with the basecaller as the equalizer.

In [17], it was observed that the error events (substitution, insertion, deletion, or no error) induced from the basecaller exhibit dependence with the previous bases and their error events. This motivated them to propose a variation of the SID that includes non-i.i.d. error events, which we denote as the SID channel *with memory*.

Channel definition. For a memory of τ bases, the input to the channel is a sequence of input bases $\{B_n\}_{n=1}^{m+\tau-1}$ that correspond to the states $\{S_\ell\}_{\ell=1}^{m}$, as introduced in Section IV. The base $B_{\ell+\tau-1}$ is corrupted by an error E_ℓ from $\mathcal{E} = \{\mathsf{Sub}, \mathsf{Ins}, \mathsf{Del}, \mathsf{NoErr}\}$. The key idea from [17] was to consider the dependence of error E_ℓ with the previous state $S_{\ell-1}$ and its error event $E_{\ell-1}$. Hence, we define the SID channel with memory by the transition probabilities $\mathbb{P}(E_\ell = e | S_{\ell-1} = s', E_{\ell-1} = e')$ for all $e, e' \in \mathcal{E}$ and for all $s' \in \Omega$. The channel output is a sequence of bases that are appended according to the synchronization errors in the input bases. There are four cases to consider:

- Substitution $E_\ell = \mathsf{Sub}$: substitute $B_{\ell+\tau-1}$ with a base from $\mathcal{B} \backslash \{B_{\ell+\tau-1}\}$ and append it to the output.
- Insertion $E_\ell = \mathsf{Ins}$: insert L copies of $B_{\ell+\tau-1}$ into the output, where L is random.
- Deletion $E_\ell = \mathsf{Del}$: no change to output.
- No error $E_\ell = \mathsf{NoErr}$: append $B_{\ell+\tau-1}$ to the output.

This channel was taken furthest to date in [18]. The main contributions included deriving a MAP decoder, which facilitated a Monte Carlo technique to estimate achievable information rates of the basecalled nanopore channel. The MAP decoder was further leveraged to construct a concatenated coding scheme based on LDPC codes. The performance of this coding scheme was particularly good when more than 5 reads of the same DNA strand were combined to improve decoding. Its single-read performance was not great, however it is common to have multiple reads in DNA storage.

C. Comparing the channel models

In this article, we have considered three channel models for nanopore sequencing. A strong case could be made for each channel, depending on which practical considerations are deemed most important. While a direct comparison is yet to be conducted in the literature, let us note a few scenarios where each of these channels could shine.

Sampling rate. It is known that increasing the min. number of samples per segment significantly improves the performance of channels with noisy duplications [15]. At some threshold, they can effectively be ignored as in the SNC that was designed for a sampling rate of roughly 50kHz, thus putting more emphasis on other distortions in the sequencer. Modern Oxford nanopore sequencers use a sampling rate of 4kHz (recently increased to 5kHz), putting a far greater importance on the noisy duplications as modelled in the NNC.

Parameter estimation. Since the NNC and SNC are based on the pore model, they require learning the channel mapping f over a potentially large state-space Ω. For the R9.4 Oxford nanopore, it was demonstrated in Section III that $\tau = 5$ was sufficient, but for the recent R10.4 Oxford nanopore it has roughly doubled to around $\tau = 10$, making accurate parameter estimation of the pore model challenging. On the other hand, learning the transition probabilities that define the SID channel with memory is relatively simple by counting the frequency of errors in the many basecalled datasets that are available.

Multiple reads. The ethos behind the NNC and SNC is to faithfully model the nanopore sequencing process for a single read such that we can achieve the fundamental limit of the nanopore channel. This can lead to large gains in rate over SID channels for a single read. For multiple reads (e.g., more than 5), it is not clear if this ethos is still justified in the face of other practical considerations like decoding complexity. Hence, there is an interesting trade-off between each model depending on how many reads are available for each codeword.

VI. The Decoding Problem

Once we have a sequence of observations from the nanopore sequencer, the next step is to detect the most likely sequence of bases that passed through the channel. It is commonplace to simplify the complexity of the detection algorithm by making a strategic hard decision in exchange for a lower time-complexity, like in traditional noisy channels, however it is important to consider how this decision degrades the channel. The estimation of state sequence S_1^m using the observations $Y_1^{T_m}$ is limited by how the segmentation T_1^m is handled in the detection algorithm. Since the states and segmentation are inherently coupled, the optimal detector must jointly consider the most likely state sequence across all possible segmentations. This translates to marginalizing the segmentation out of the joint probability $\mathbb{P}(Y_1^{T_m}, T_1^m, S_1^m)$ to get $\mathbb{P}(Y_1^{T_m}, S_1^m)$, which can be used to estimate S_1^m using traditional techniques. Since this marginalization process satisfies the global constraint of m segments, we refer to this approach generally as "global segmentation".

Given that the marginalization is an expensive process, it is tempting to make a hard decision in order to simplify its complexity. A significant reduction can be achieved by performing segmentation locally, that is, decoding without satisfying the constraint of m segments. We refer to this approach generally as "local segmentation". The result of the local segmentation approach is that the duplication errors become insertions and deletions of bases. In fact, this approach was taken by the equalization algorithms in the two channel models discussed in Section V, which both considered insertion and deletion errors. Alternatively, the class of segmentation algorithms [32] strike a sensible balance between complexity and accuracy by making a hard decision on the segmentation but still satisfying the global constraint of m segments.

Firstly, let us understand the marginalization process in more detail. After that, we will discuss segmentation

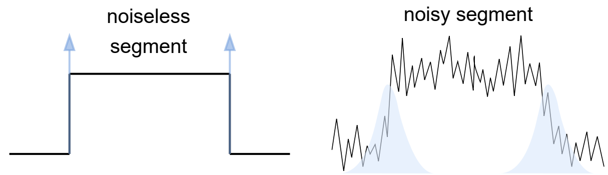

Fig. 6. Comparing a noiseless segment with a noisy segment. Blue represents the uncertainty in where the segment starts and ends.

algorithms and their implications on coding. Finally, we discuss a few important problems in the literature that relate to the decoding problem.

A. Marginalizing the segmentation

The process of marginalizing the segmentation out of the received observations is to transform it into a statistic that no longer considers the segmentation. This may be described as the equalization of duplications in the observations of the channel output, leaving behind the induced state information for further processing.

The marginalization process considers the segmentation likelihoods

$$L(y_1^{t_m}, s_1^m | t_1^m) = \mathbb{P}(Y_1^{T_m} = y_1^{t_m}, S_1^m = s_1^m | T_1^m = t_1^m)$$

for all possible segmentations t_1^m. By summing over all possible segmentations, weighted by their probability $\mathbb{P}(T_1^m = t_1^m)$, we have the marginalized probability

$$\mathbb{P}(Y_1^{T_m} = y_1^{t_m}, S_1^m = s_1^m) = \sum_{t_1^m} L(y_1^{t_m}, s_1^m | t_1^m) \mathbb{P}(T_1^m = t_1^m).$$

Fortunately, the sequential nature of the segments in the segmentation allows an efficient computation using dynamic programming [33].

The segmentation likelihood $L(y_1^{t_m}, s_1^m | t_1^m)$ gives insight into the synchronization of the channel symbols. Ideally, we want any segmentation that is not the correct one to have a high penalty (i.e., $L(y_1^{t_m}, s_1^m | t_1^m) = 0$) so that the total contribution is determined only by the correct segmentation T_1^m. In practice, the noisy levels mean that there are many segmentations with a significant contribution, which makes perfect segmentation more challenging due to the increased uncertainty. This is visualized in Fig. 6 by comparing a noiseless segment with a noisy segment, where the uncertainty in the jump times are in illustrated in blue. For the noiseless segment, it is clearly very easy to do segmentation since there is a single obvious candidate, and hence the segmentation likelihood is effectively a delta distribution that has all of its mass in one segmentation. For the noisy segment, it is no longer trivial to do segmentation since the segmentation likelihood is spread out, and therefore we must consider many possible segmentations to estimate the level of the segment.

B. Segmentation algorithms

For state-estimation detection algorithms, we saw that the complexity of detection was greatly influenced by the requirement to marginalize over all of the many possible segmentations. For detection algorithms based on segmentation algorithms, the key idea is to make a hard decision on the segmentation T_1^m by choosing the most likely one jointly with the most likely state sequence S_1^m. Conditioned on a state sequence s_1^m, the segmentation algorithm finds the best way to split the observations $y_1^{t_m}$ into a series of m segments, where m is the number of inputs that is known a priori. That is, we choose the segmentation $\mathbf{t}^*_{s_1^m}$ that achieves

$$d(y_1^{t_m}, s_1^m) = \min_{t_1^m} \left[-\log L(y_1^{t_m}, s_1^m | t_1^m) \mathbb{P}(T_1^m = t_1^m) \right].$$

Then, the MAP state-segmentation estimate is the pair $(\mathbf{s}^*, \mathbf{t}^*)$ that achieves $\min_{s_1^m} d(y_1^{t_m}, s_1^m)$.

The estimate $(\mathbf{s}^*, \mathbf{t}^*)$ can be computed efficiently when the channel input is Markov and the duplications are i.i.d. using the generalized Viterbi algorithm (GVA) [15]. This detector is worse than the optimal detector that performs marginalization since it ignores all but a single segmentation, which in general does not hold all of the information in the segmentation. However, the benefit is that we avoid the expensive process of marginalizing out the segmentation that is now part of the estimate. In algorithmic terms, we are changing from a summation to a maximization, of which the latter has many good approximations that have been used successfully in the classical Viterbi decoder. In addition, this decoder makes it possible to formulate a distance that can be used as a design criterion for coding schemes.

DTW distance. There is a vast body of literature on segmentation algorithms that include some very efficient implementations [32], and therefore it is natural to ask whether any of these algorithms can serve as a close approximation to the distance $d(y_1^{t_m}, s_1^m)$. One such algorithm is DTW [34] that has been extensively studied by computer scientists. This algorithm computes a distance measure between two signals that have been non-linearly warped (stretched or contracted in various regions of the signal) and attempts to warp the signals such that the cost of alignment is minimized.

The DTW distance between channel observations $\mathbf{y} = y_1^{t_m}$ and levels $\mathbf{x} = (f(s_\ell))_{\ell=1}^m$ of the state sequence $\mathbf{s} = s_1^m$ is of the form

$$d_{\mathsf{DTW}}(\mathbf{y}, \mathbf{x}) = \min_{\psi,\phi} ||\psi(\mathbf{y}) - \phi(\mathbf{x})||^2$$

where ψ and ϕ are *warping functions*. DTW jointly chooses the warping function ϕ to warp $\mathbf{x}$ using sample duplications and ψ using sample deletions such that the misalignment cost, measured as Euclidean distance squared, is minimized. For channels with only sample duplications, the warping function ψ can be ignored since it is always the identity function $\psi(\mathbf{y}) = \mathbf{y}$, but we include it here since it is required for symmetry. This minimization can be efficiently computed in $O(mT_m)$ time using the DTW algorithm that is based on a dynamic programming formulation [34].

The DTW distance d_{DTW} for pairs of arbitrary-length sequences over the real numbers do not form a metric space. For any $\mathbf{x}, \mathbf{y}, \mathbf{z}$:

1) $d_{\mathsf{DTW}}(\mathbf{x}, \mathbf{x}) = 0$.
2) If $\mathbf{x} \neq \mathbf{y}$, then $d_{\mathsf{DTW}}(\mathbf{x}, \mathbf{y}) \geq 0$.
3) $d_{\mathsf{DTW}}(\mathbf{x}, \mathbf{y}) = d_{\mathsf{DTW}}(\mathbf{y}, \mathbf{x})$.
4) The triangle inequality does not hold.

where 2) and 4) are not properties of a metric space. For example, if we have $\mathbf{x} = (0, 1, 1, 2)$ and $\mathbf{y} = (0, 1, 2)$, and $\mathbf{z} = (0, 2, 2)$. The warping paths for $\mathbf{x}$ and $\mathbf{y}$ that achieve the DTW distance are $\phi(\mathbf{x}) = \mathbf{y}$ and $\psi(\mathbf{y}) = (0, 1, 1, 2)$ (that is, only sample duplications), giving $d_{\mathsf{DTW}}(\mathbf{y}, \mathbf{x}) = 0$. This is an example of the equality in Property 2. Further, we have $d_{\mathsf{DTW}}(\mathbf{y}, \mathbf{z}) = 1$ and $d_{\mathsf{DTW}}(\mathbf{x}, \mathbf{z}) = 2$. Since $d_{\mathsf{DTW}}(\mathbf{x}, \mathbf{z}) > d_{\mathsf{DTW}}(\mathbf{x}, \mathbf{y}) + d_{\mathsf{DTW}}(\mathbf{y}, \mathbf{z})$, this is an example of the triangle inequality not holding as in Property 4.

For many duplication channels, the DTW distance is not quite the optimal, that is, it only approximates the optimal distance d. In [35], it was proven that segmentation algorithms like DTW are only optimal for joint channel distributions that are log-additive. The Euclidean distance metric used in DTW further restricts the channel and the input: the duplications must be i.i.d. geometrically distributed, the noise must be AWGN, and the input codewords must be uniformly distributed. In practice, the duplications do not greatly influence d, thus DTW can still be a good approximation for other channels.

Code design based on DTW. In [36], the DTW distance was used to derive a union bound on the pairwise error probability between codewords for a class of noisy duplication with geometric-like characteristics, including discrete uniform distributions as a special case. The exponent of the error bound is computed by an algorithm in $O(m^2)$ time (with a constant depending on the support of the duplications) for codeword length m. It is currently limited to short block lengths in practice, but it is nonetheless a promising technique for code design.

A notable property of d_{DTW} is that the distance between adjacent segment levels is crucial for accurate segmentation. Hence, a useful design criterion for improving synchronization is the *jumping distance* $d_{\mathtt{J}}(s, s') = |f(s) - f(s')|$ for states $s, s' \in \Omega$ with non-zero transition probabilities. This relates to the observation in Fig. 6 that more noise makes segmentation more difficult. Conversely, if we design a codebook such that the minimum $d_{\mathtt{J}}(s, s')$ is high, then synchronization of the symbols in each codeword will be easier. This criterion is in the spirit of constraint codes that improve the channel characteristics, e.g., run-length constraint codes in magnetic recording.

C. Neural network basecallers

Current-day DNA storage systems are almost exclusively based on NN basecallers that attempt to estimate the bases that passed through the nanopore. These basecallers are developed for *genomic sequencing* by training on variable-length strands of DNA found in nature, specifically the genomes of various target species. This differs from the DNA storage application where there are exactly m inputs encoded into fixed-length synthetic strands of DNA, and often requires embracing the local segmentation approach. This casts some doubt over their direct application to DNA storage, despite their success in genomic sequencing.

A solution to this problem was proposed in [37]. Instead of making hard decisions on the base sequence, the likelihood estimates from inside the network were used for additional processing. For so-called "flip-flop" basecallers, these likelihoods correspond to transition probabilities between "stay" and "go" states for every sample, where "stay" is to duplicate the current base and "go" is to transition to a different base (beginning at a new segment). It is possible to enforce the segment constraint by employing a second stage of decoding. This is a promising technique for minimizing channel degradation while still taking advantage of NN models.

D. Short reads vs long reads

The length of the DNA strands processed by Oxford nanopore sequencers can be very long (up to $10^3 - 10^5$) in comparison to other sequencing technologies (up to $10^2 - 10^3$). This reduces the number of reads for long files stored in multiple strands of DNA in the solution. The state-of-the-art DNA storage systems use *shotgun sequencing*, which is to randomly sample and read strands from the pool. This approach is formally described by the *coupon collector's problem* that, in our case, is concerned with the average number of DNA strands to be read in order to reconstruct the entire file. On the other hand, long reads are more sensitive to synchronization errors and this needs to be compensated by reducing the storage efficiency (bits/nt) in order to maintain the desired error rate.

E. Short dwell times and homopolymer stretches

After basecalling, higher error rates are often attributed to short dwell times and homopolymer stretches (consecutive repeated bases) [38]. These properties can be explained using properties of general duplication channels that include the nanopore channel.

Much like long reads, shorter dwell times increase the difficulty of synchronization since there are fewer samples per current level, resulting in a reduction of storage efficiency. This was leveraged in [16] to model the average current levels perfectly in the presence of a high sampling rate, since this resulted in a lot of samples per level. The same property was discussed in [15], where it was observed that decreasing minimum duration of the NNC significantly reduced storage efficiency.

The effect of homopolymer stretches can be explained using a noiseless duplication channel, better known as the sticky channel. Sticky channels transmit information entirely in the duration of consecutive identical symbols, and often have a low channel capacity [28]. Hence, it is expected that codewords with more information encoded in homopolymers result in higher error rates unless there is sufficient redundancy to compensate. However, it is important to note that there are additional motivations for limiting homopolymer stretches, in particular due to the synthesis process that occurs before sequencing.

VII. Open Problems

In this article, we explored the nanopore sequencing process in the context of DNA storage. Given that nanopore theory is still early in its development from an information-theoretic perspective, numerous open problems were raised throughout the article. Many of them surround the relatively unexplored noisy duplication channels. Let us briefly summarize and further discuss some of the open problems that may be interesting to the information theory community.

Capacity. From simulation results, it appears the Markov-constrained capacity is capable of rates near capacity at least in the low-noise regime. However, this observation is yet to be proven for noisy duplication channels as it has been for finite-state channels. In addition, there is not an algorithm that optimizes Markov sources for noisy duplication channels to the Markov-constrained capacity. Steps toward this goal were taken in [33], which demonstrated how the generalized Blahut-Arimoto algorithm [39] could be used to optimize achievable rates, however it likely does not reach capacity.

Coding. With recent advancements on efficient codes for synchronization channels, there is an opportunity to apply these to noisy duplication channels. The coding problem has only been partially addressed in [18], [33], [36] for the idealized channel, and in [37] using hybrid decoders for the nanopore channel. However, additional challenges are present in the context of DNA storage due to other processes alongside the sequencing process. This often manifests as source constraints to the channel code, e.g., the minimum run-length constraint and the GC balancing constraint. This is a hot topic in the literature on algebraic coding (e.g., [40]–[42]).

Pore model estimation. As discussed in Section III, there is currently no estimation algorithm for learning the channel parameters of a noisy duplication channel with theoretical guarantees. At present, we often rely on the pore models that are released by ONT (e.g., the latest pores in [20], and the older pores in Scrappie [22]). The accuracy of these models is not clear. To compound this issue, the parameter estimation problem is a greater challenge for the latest R10.4 pore, which has doubled τ from 5 to 10 for an accurate pore model (approx. one million nanopore states). This could force the development of models with a reduced τ and the resulting model inaccuracy included as noise, e.g., using the fading model from the SNC in [16].

While not covered in this article, there are additional open problems due to other processes in a DNA storage system that are handled by higher layer protocols. These problems include *random access* [4], [11], [13], i.e., the retrieval of files stored in pools of DNA strands, and *trace reconstruction* [9], [43], i.e., the decoding of files using multiple reads from the nanopore channel.

In conclusion, this article has highlighted the relatively unexplored field of nanopore sequencing channels in the context of DNA storage, raising numerous open problems from an information-theoretic perspective. These open problems present intriguing opportunities for the information theory community to further investigate and develop innovative solutions, with the end goal of unlocking the full potential of DNA storage systems with nanopore sequencing as the reading process. The exploration of these challenges is crucial to the pursuit of deepening our understanding in this exciting area of research.

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